

Reyhaneh (Rey) Kharazmi Pour

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Research Interests

Online algorithms and optimization; networked decision-making; strategic interactions; resource allocation under uncertainty.

Education

University of California, Irvine

2023 – Present

Ph.D., Operations, Decisions, and Technology Management | GPA: 4.0/4.0

- University Fellowship (2023–2028); Graduate Dean’s Recruitment Fellowship
- Teaching Assistant: Management Science Analytics; Operations Management; Statistics for Business Decision Making

Sharif University of Technology

2017 – 2022

B.Sc., Industrial Engineering | GPA: 4.0/4.0 (final two years)

- Thesis: *Optimal Referral Marketing with a Simulation-Optimization Approach* [\[link\]](#)
- Teaching Assistant: Project Control; Transportation Planning; Facilities Planning

Research

Learning to Monetize Influence: Online Ad Allocation with Edge-Aware Dual Mirror Descent

Working Paper

with Prof. John Turner

Studies online ad allocation in social networks where impression value depends on time-sensitive complementarities across connected users. Develops a network-aware online algorithm based on dual mirror descent that incorporates expected future coordination value under budget constraints.

Regulating Farmland Abandonment under Network Cost Spillovers

Working Paper

with Prof. Luyi Gui

Analyzes farmland abandonment in agricultural markets with network-dependent costs. Develops an equilibrium model to study how penalties and subsidies affect participation, welfare, and the emergence of exit cascades.

Optimal Paid Referral Marketing

B.Sc. Thesis

with Prof. Erfan HassanNayebi

Designed a simulation-optimization framework for referral campaigns under budget constraints; combined agent-based simulation with optimization.

Industry Experience

Business Data Analyst, Digikala

2021 – 2023

Tehran, Iran | Largest e-commerce platform in Iran

- Designed and analyzed controlled experiments (A/B tests) to evaluate product and pricing strategies at scale
- Developed data-driven frameworks for customer segmentation and lifecycle modeling to support targeted decision-making
- Analyzed large-scale operational and behavioral data to identify inefficiencies, improve pricing policies, and detect anomalous patterns in promotional usage

Additional Work

Influencer Seeding under Budget Constraints

[\[link\]](#)

Modeled influence maximization using network centrality measures and simulated budget-constrained seeding strategies.

ToxiClassify: Multi-Label Toxic Comment Classification

[\[link\]](#)

Built NLP models including CNNs and transformer-based architectures for multi-label classification.

Truck Scheduling via Mixed-Integer Linear Programming

Formulated and solved a MILP for truck sequencing under capacity and precedence constraints.

Technical Skills

Programming	Python, SQL
Optimization	Gurobi, CPLEX, AMPL; linear and integer programming
ML / Statistics	regression, classification, clustering; scikit-learn, XGBoost

Presentations

UCI Research Festival

May 2025

University of California, Irvine

Friendfluence: A Near-Optimal Algorithm Harnessing Social Connections in Ad Assignment

INFORMS Annual Meeting

Oct 2025

Atlanta, GA

Ad Assignment with Social Influence: A Scalable Optimization Approach